Mobile Application Development – CA3: Review 1

Idea & AI Use-Case Proposal

Milestone: Problem definition, AI/ML use case, and project scope

Section	Details
Group Number	*18*
Team Members	*Prachi Priya 25MCAR0111 , Suvendu Mata 25MCAR0136*
Title	*CraveReels – A Video-First Food Discovery and Smart Price Comparison Application*
App Name	*CraveReels*
Proposed AI/ML Feature	*Google Gemini AI-powered Food Advisor for food recommendations, craving-based suggestions, and food delivery platform deal comparisons. The application also supports rule-based recommendations using user preferences.*
Why AI is Needed	*AI helps users discover suitable dishes based on their mood, cravings, budget, and vegetarian preferences. It can generate food suggestions and compare ordering options across platforms, helping users make informed food choices.*
Target Users	*Students, working professionals, food lovers, and users looking for personalised food recommendations and budget-friendly meal options.*
Add-ons as per Remarks	*The project includes an AI Food Advisor and a food delivery deal comparison feature using Google Gemini AI.*

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Mobile Application Development – CA3: Review 2

AI/ML Model & Data Review

Milestone: Dataset preparation, model selection, and training approach

Section	Details
Group Number	*18*
Team Members	*Prachi Priya 25MCAR0111 , Suvendu Mata 25MCAR0136*
Title	*CraveReels – A Video-First Food Discovery and Smart Price Comparison Application*
App Name	*CraveReels*
Dataset Used	*The current AI implementation uses user-entered prompts, food preferences, and currently viewed dish information as input. No dedicated machine-learning training dataset is identified in the reviewed backend code*
Data Preprocessing	*The backend validates the user's text prompt and adds available contextual information, including mood, craving, budget, vegetarian preference, dish name, restaurant, and price, before sending the request to Gemini AI*
Model / Algorithm Chosen	*Google Gemini AI through the @google/genai SDK. The backend uses the generateContent method to generate food recommendations and platform comparisons. Rule-based recommendation logic is also described in the project metadata.*
Training & Validation Approach	*The current implementation uses a pre-trained Gemini model through an API rather than training a custom machine-learning model. Formal train-test splitting and model validation metrics have not been identified in the reviewed files.*
Model Performance (so far)	*The Gemini API integration and response-handling logic are present in the backend. Accuracy, precision, recall, and other quantitative performance metrics have not yet been established.*
Integration Plan	*The frontend sends a request to the backend endpoint /api/gemini/food-compare. The backend uses Google Gemini AI and returns the generated response to the application. The API key is configured through the GEMINI_API_KEY environment variable.*
Add-ons as per Remarks	*The backend includes input validation, error handling, and a fallback response when the Gemini API key is unavailable*

Evaluation – AI Integration & Performance (partial)

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Mobile Application Development – CA3: Review 3

App Development & Functionality Review

Milestone: Core app build and functional integration of the AI feature

Section	Details
Group Number	*18*
Team Members	*Prachi Priya 25MCAR0111 , Suvendu Mata 25MCAR0136*
Title	*CraveReels – A Video-First Food Discovery and Smart Price Comparison Application*
App Name	*CraveReels*
Features Implemented	*The project is designed for short-form food discovery, craving-based recommendations, and delivery platform deal comparisons. The backend includes an AI Food Advisor endpoint, contextual prompt processing, and error handling. The working status of each frontend feature requires testing.*
Tech Stack Used	• *Frontend: React, TypeScript, Vite, and Tailwind CSS.* • *Backend: Node.js and Express.js.* • *AI Integration: Google Gemini AI using the @google/genai SDK.* • *Configuration: Environment variables using dotenv.*
AI Feature Functionality	*The user provides a food-related query. The backend combines the query with available user preferences and current dish information. Gemini AI processes the contextual prompt and generates food recommendations and delivery platform comparisons. The response is returned to the frontend*
Error Handling / Edge Cases	*The backend validates the prompt and returns an error for invalid input. It provides a fallback response when the Gemini API key is unavailable and returns an error response if the API request fails.*
Known Issues / Bugs	*The frontend connection to the AI endpoint, API key configuration, and actual AI response generation need to be tested before confirming complete functionality. Generated prices, offers, and delivery times also require independent verification.*
Add-ons as per Remarks	*Test the AI endpoint with different food queries, budgets, cravings, and dietary preferences. Verify the generated responses and handle API failures appropriately.*

Evaluation – Application Functionality

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Mobile Application Development – CA3: Review 4

UI/UX & AI Output Presentation Review

Milestone: Interface design and clarity of AI-generated results

Section	Details
Group Number	*18*
Team Members	*Prachi Priya 25MCAR0111 , Suvendu Mata 25MCAR0136*
Title	*CraveReels – A Video-First Food Discovery and Smart Price Comparison Application*
App Name	*CraveReels*
Screens Designed	*The application is intended to provide food discovery and AI-assisted recommendation experiences. The exact screens available in the current frontend should be confirmed by running the application*
Navigation Flow	*Users explore food content and provide a food-related query or preferences to access AI-assisted recommendations. The AI response is retrieved through the backend endpoint and displayed in the application if the frontend integration is implemented.*
AI Output Presentation	*The backend is instructed to generate structured recommendations, identify best food options, distinguish premium bestsellers from budget-friendly choices, and compare delivery platforms. The final presentation depends on the frontend implementation.*
Interactive Components Used	*The project uses React and Lucide icons. The specific AI input fields, buttons, lists, and other interactive components must be verified in the frontend source code.*
Responsiveness Check	*Test the application on different mobile screen sizes and desktop browsers. Record the devices and screen resolutions actually tested.*
Add-ons as per Remarks	*Improve the readability of AI recommendations, highlight budget-friendly choices, and ensure that the interface handles loading states and API errors*

Evaluation – UI/UX Design

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Mobile Application Development – CA3: Review 5

Final Deployment & Demo Review

Milestone: Deployed application, final AI performance, and live demo

Section	Details
Group Number	*18*
Team Members	*Prachi Priya 25MCAR0111 , Suvendu Mata 25MCAR0136*
Title	*CraveReels – A Video-First Food Discovery and Smart Price Comparison Application*
App Name	*CreaveReels*
Deployment Details	*The project can be run locally using Node.js and the command npm run dev. Configure the GEMINI_API_KEY environment variable before testing live Gemini API requests. Final hosting and deployment details are to be added after deployment.*
Final AI Model Performance	*The application integrates a pre-trained Gemini AI model for food-related recommendations and delivery platform comparisons. Final performance results must be recorded after testing the application with representative queries. No verified accuracy metrics are currently available.*
Final Feature Summary	*CraveReels combines short-form food discovery with AI-assisted food recommendations and delivery platform comparisons. The AI feature uses user queries and available preferences to generate contextual suggestions. The application also includes a backend fallback response and error handling.*
Demo Notes	1. *Launch the application.* 2. *Open the AI Food Advisor, if available in the frontend.* 3. *Enter a query such as "Suggest a filling meal under ₹200."* 4. *Test different cravings, budgets, and vegetarian preferences.* 5. *Demonstrate the generated food recommendations and delivery platform comparison.* 6. *Show how the application handles missing API configuration or failed requests.*
Project Report Submitted	*Yes*
Add-ons as per Remarks	*Complete end-to-end testing, verify the accuracy of displayed offers, resolve identified bugs, and document the final deployment details before submission*

Final Evaluation – CA3 Rubric (Total 15 Marks)

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